

LEARNING EQUIVARIANT NON-LOCAL ELECTRON DENSITY FUNCTIONALS

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ABSTRACT

The accuracy of density functional theory hinges on the approximation of *non-local* contributions to the exchange-correlation (XC) functional. To date, machine-learned and human-designed approximations suffer from insufficient accuracy, limited scalability, or dependence on costly reference data. To address these issues, we introduce **Equivariant Graph Exchange Correlation (EG-XC)**, a novel non-local XC functional based on equivariant graph neural networks (GNNs). Where previous works relied on semi-local functionals or fixed-size descriptors of the density, we compress the electron density into an SO(3)-equivariant nuclei-centered point cloud representation for efficient non-local atomic-range interactions. By applying an equivariant GNN on this point cloud, we capture molecular-range interactions in a scalable and accurate manner. To train EG-XC, we differentiate through a self-consistent field solver requiring only energy targets. In our empirical evaluation, we find EG-XC to accurately reconstruct “gold-standard” CCSD(T) energies on MD17. On out-of-distribution conformations of 3BPA, EG-XC reduces the relative MAE by 35% to 50%. Remarkably, EG-XC excels in data efficiency and molecular size extrapolation on QM9, matching force fields trained on 5 times more and larger molecules. On identical training sets, EG-XC yields on average 51% lower MAEs.

1 INTRODUCTION

Kohn-Sham Density Functional Theory (KS-DFT) is the backbone of computational material and drug discovery (Jones, 2015). It is a quantum mechanical method to approximate the ground state energy of an N_e -electron system by finding the electron density $\rho \in \mathcal{D}_{N_e} = \{\rho: \mathbb{R}^3 \rightarrow \mathbb{R}_+, \int_{\mathbb{R}^3} \rho(r) dr = N_e\}$ that minimizes the energy functional $E: \mathcal{D}_{N_e} \rightarrow \mathbb{R}$. This functional maps electron densities to energies and is composed of

$$E[\rho] = T[\rho] + V_{\text{ext}}[\rho] + V_{\text{H}}[\rho] + E_{\text{XC}}[\rho] \quad (1)$$

$T: \mathcal{D}_{N_e} \rightarrow \mathbb{R}$ is the kinetic energy functional, $V_{\text{ext}}: \mathcal{D}_{N_e} \rightarrow \mathbb{R}$ the external potential due to positively charged nuclei, $V_{\text{H}}: \mathcal{D}_{N_e} \rightarrow \mathbb{R}_+$ the Coulomb energy between electrons, and, finally, $E_{\text{XC}}: \mathcal{D}_{N_e} \rightarrow \mathbb{R}_-$ the *exchange-correlation* (XC) functional (Cramer, 2004). While T, V_{ext} and V_{H} permit analytical computations, the exact form of E_{XC} remains unknown, and its approximation frequently dominates DFT’s error (Kim et al., 2013).

Machine learning has emerged as a promising data-driven approach to approximate E_{XC} (Kulik et al., 2022; Zhang et al., 2023). Many such ML functionals adopt the classical approach and define E_{XC} as the integral of a learnable XC energy density $\epsilon_{\text{XC}}: \mathbb{R}^{d_{\text{basis}}} \rightarrow \mathbb{R}$ (Perdew, 2001):

$$E_{\text{XC}}[\rho] = \int_{\mathbb{R}^3} \rho(r) \epsilon_{\text{XC}}(\mathbf{g}(r)) dr \quad (2)$$

where $\mathbf{g}: \mathbb{R}^3 \rightarrow \mathbb{R}^d$ are properties of the electron density ρ (Dick & Fernandez-Serra, 2021; Nagai et al., 2022). If $\mathbf{g}$ only depends on density quantities from its infinitesimal neighborhood, e.g., $\rho(r), \nabla \rho(r), \dots$, the resulting functionals are called *semi-local*. While this integrates well into existing quantum chemistry code, non-local interactions like *Van der Waals* forces exceed the functional class (Kaplan et al., 2023). In contrast, *non-local* functionals can capture such interactions by depending on multiple points in space simultaneously, e.g., $E_{\text{XC}}[\rho] = \iint_{\mathbb{R}^3} \rho(r) \rho(r') \epsilon_{\text{XC}}(\mathbf{g}(r), \mathbf{g}(r')) dr dr'$.

Non-locality can be further separated into *atomic-range* interactions, which depend on the electron density around the nucleus, and *molecular-range* interactions, which depend on the electron density within typical molecular length scales. But, existing non-local functionals either scale poorly computationally (Zhou et al., 2019) or require costly reference data (Margraf & Reuter, 2021; Bystrom & Kozinsky, 2022). The critical challenge in XC functionals lies in efficiently capturing non-local molecular-range interactions at scale.

To this end, we propose to leverage the success of equivariant graph neural networks (GNNs) to learn non-local XC functionals through our novel **Equivariant Graph Exchange Correlation (EG-XC)**¹. To enable computationally efficient non-local interactions, we propose two key innovations: (1) We compress the electron density into an SO(3)-equivariant atomic-range point cloud representation by convolving the electron density with an SO(3)-equivariant kernel at the nuclear positions. (2) We use SO(3)-equivariant GNNs on this point cloud representation to efficiently capture molecular-range information. Finally, we use the embeddings to define a non-local feature density $\mathbf{g}_{\text{NL}}: \mathbb{R}^3 \rightarrow \mathbb{R}^d$ for the XC energy density ϵ_{XC} from Equation 2. Compared to previous approaches, our finite point cloud embeddings are neither dependent on the nuclear charge nor the basis set but purely derived from the density ρ . To train on energies alone, we differentiate through the minimization of Equation 1 (Li et al., 2021).

In our experimental evaluation, EG-XC improves upon the learnable semi-local XC-functional on molecular dynamic trajectories, extrapolation to both out-of-distribution conformations and increasingly larger molecules. In particular, we find EG-XC to reduce errors of the semi-local functional by a factor of 2 to 3, on par with accurate ML force fields combined with DFT calculations (Δ -ML). In extrapolation to unseen structures, we find EG-XC to accurately reproduce out-of-distribution potential energy surfaces unlike force fields (incl. Δ -ML) while reducing the MAE by 35% to 50% compared to the next-best tested method. Finally, we find EG-XC to have an excellent data efficiency, achieving similar accuracies to the best force fields with 5 times less data. To summarize, we demonstrate that GNN-driven functionals push the frontier of non-local XC functionals and provide a promising path towards accurate and scalable DFT calculations.

2 BACKGROUND

Notation. To denote functionals, i.e., functions mapping from functions to scalars, we write $F[f]$ where F is the functional and f the function. We use superscripts in brackets⁽ⁱ⁾ to indicate sequences and regular superscripts of bold vectors, e.g., $\mathbf{x}^i$, to indicate the i -th irreducible representation of the SO(3) group. We generally use $r \in \mathbb{R}^3$ to denote points in the 3D Euclidean space. A notable exception is the finite set of nuclei positions $\{R_1, \dots, R_{N_n}\}$ for which we will use capital letters. The norm of a vector is given by $\|r\| = \|r\|_2$. For directions, i.e., unit length vectors, we use $\hat{r} = \frac{r}{\|r\|}$.

Kohn-Sham density functional theory is the foundation of our work. Here, we provide a very brief introduction. For more details, we refer the reader to Appendix A or Lehtola et al. (2020). In KS-DFT, the electron density ρ is represented by a set of orthogonal orbitals $\phi_i: \mathbb{R}^3 \rightarrow \mathbb{R}$, $\phi_i(r) = C_i^T \chi(r)$, that are defined as linear combinations of a basis set of atomic orbitals $\chi_a: \mathbb{R}^3 \rightarrow \mathbb{R}$:

$$\rho(r) = \phi(r)^T \phi(r) = \chi(r)^T C C^T \chi(r) = \chi(r)^T P \chi(r) \quad (3)$$

where $P = C C^T \in \mathbb{R}^{N_{\text{basis}} \times N_{\text{basis}}}$ is the so-called density matrix. Given the representation in a finite basis set, one can compute the kinetic energy T , external potential V_{ext} , and electron-electron repulsion energies V_{H} analytically. Unfortunately, such analytical expressions are generally unavailable for $E_{\text{XC}}[\rho]$ as in Equation 2. Thus, one relies on numerical integration (Lehtola et al., 2020).

To minimize Equation 1, one typically uses the self-consistent field (SCF) method that we outline in greater detail in Appendix B. The SCF method is an iterative two-step optimization where one first computes the so-called Fock matrix $F = \frac{\delta E}{\delta \phi}$ based on the current coefficients C . Then, the optimal coefficients $C \in \mathbb{R}^{N_{\text{basis}} \times N_e}$ are obtained by solving the generalized eigenvalue problem

$$F C = S C E \quad (4)$$

with $S_{\mu\nu} = \int_{\mathbb{R}^3} \chi_\mu(r) \chi_\nu(r) dr$ being the overlap matrix of the atomic orbitals and E being a diagonal matrix. The procedure is repeated until convergence of P .

¹We provide the source code for review on <https://higshare.com/s/656a3c16b0b1e881095>

Equivariance allows defining symmetries of functions. A function $f: \mathcal{X} \rightarrow \mathcal{Y}$ is equivariant to a group G iff $\forall g \in G, x \in \mathcal{X}, f(G_g^X x) = G_g^Y f(x)$ where G_g^X, G_g^Y are the representations of g in the domain $\mathcal{X}$ and codomain $\mathcal{Y}$, respectively. Invariance is a special case of equivariance where $G_g^Y = 1$, i.e., $f(G_g^X x) = f(x)$, for all $g \in G, x \in \mathcal{X}$.

The real spherical harmonics $Y^l: \mathbb{R}^3 \rightarrow \mathbb{R}^{2l+1}, l \in \mathbb{N}_+$, are a well-known example of SO(3)-equivariant functions as they transform under rotation according to the Wigner D matrices $D_R^l \in \mathbb{R}^{(2l+1) \times (2l+1)}, \forall R \in \text{SO}(3)$, with $D_R^0 = 1$ being the identity and D_R^l being 3D-rotation matrices:

$$Y^l(D_R^l x) = D_R^l Y^l(x). \quad (5)$$

Such $2l+1$ -dimensional equivariant functions $h_l: \mathbb{R}^3 \rightarrow \mathbb{R}^{2l+1}$, e.g., $h_l = Y^l$, can be recombined using the tensor product

$$(h_1 \otimes h_2)_{m_1, m_2}^l = \sum_{l_1, l_2} \sum_{m_1, m_2} C_{l_1, l_2, l}^{m_1, m_2, m} h_{l_1}^{m_1} h_{l_2}^{m_2}, \quad (6)$$

where $C \in \mathbb{R}^{(2l_1+1) \times (2l_2+1) \times (2l+1)}$ are the Clebsch-Gordan coefficients. As the Wigner D-matrices are orthogonal, it follows that the inner product of two l -equivariant functions yields an invariant one

$$h_l^T(D_R^l r)^T h_l^T(D_R^l r) = h_l^T(r)^T D_R^{lT} D_R^l h_l(r) = h_l^T(r)^T h_l(r). \quad (7)$$

While, we are interested in $E(3)$ -invariant functions, using such equivariant intermediate representations improves accuracy, expands the function class, and improves data efficiency (Schütt et al., 2017; Gastreiger et al., 2021; Batatia et al., 2022).

3 RELATED WORK

Learnable density functional approximations (DFA) of the XC functional have a long history. In their seminal paper, Kohn & Sham (1965) proposed an ϵ_{XC} fitted to a reference calculation. In this first parameterization ϵ_{XC} only depends on $g(r) = \rho(r)$. To improve accuracy, the density features g have been successively refined by including gradients and physical quantities such as the kinetic energy density $\tau(r) = \frac{1}{2} \sum_a |\nabla \phi_a(r)|^2$. The resulting *semi-local* functionals are also known as meta-GGAs with $g(r) = g_{\text{meta-GGA}}(r) := [\rho(r), \nabla \rho(r), \tau(r)]$ (Perdew, 2001). Many ML functionals use this parameterization with ϵ_{XC} frequently being a product of an MLP with a classical functional (Nagai et al., 2022; Kulik et al., 2022; Zhang et al., 2023). An appealing aspect of learning semi-local energy densities rather than $E_{\text{XC}}[\rho]$ directly is the known implementation of physical constraints (Sun et al., 2015; Kaplan et al., 2023), which inspired works by Dick & Fernandez-Serra (2021); Nagai et al. (2022). While augmenting $g_{\text{meta-GGA}}$ with hand-crafted non-local features improves the accuracy, this comes at the cost of exact constraints (Nagai et al., 2020; Kirkpatrick et al., 2021). Similarly, such physical biases are absent in kernel methods Margraf & Reuter (2021); Bystrom & Kozinsky (2022). Additionally, these methods require reference densities that are rarely available (Szabo & Ostlund, 2012). Alternative grid-based CNNs converge slowly in the number of grid points compared to specialized spherical integration grids (Zhou et al., 2019; Treutler & Ahlrichs, 1995). Lastly, machine-learned functionals have been used in other DFT contexts as well, e.g., for energy corrections on SCF-converged energies (Chen & Yang, 2024) or to model the kinetic energy functional in orbital-free DFT (Snayder et al., 2012; Mi et al., 2023; Zhang et al., 2024). Here, we propose to extend semi-local models with expressive fully learnable non-local features obtained from standard integration grids. This allows us to leverage the physical biases of meta-GGAs while allowing efficient non-local interactions with fast integration. Although this work focuses on the XC functional, the methods we present can be transferred to these applications as well.

Machine learning force fields have long functioned as a cheap but inaccurate alternative to quantum mechanical calculations by directly approximating the potential energy surface from data without solving the electronic structure problem (Unke et al., 2021). Contemporary force fields generally rely on graph neural networks (GNNs) (Schütt et al., 2017) representing molecules in terms of graphs with atoms as nodes. The advent of SO(3)-equivariant models led to significant improvements in accuracy and data efficiency, closing the gap to full-fidelity quantum mechanical calculations (Batzner et al., 2022; Batatia et al., 2022). While they effectively aim to accomplish the same goal as DFT at a fraction of the cost, their out-of-distribution accuracy is a common problem (Stocker et al., 2022). In EG-XC, we transfer the success of equivariant GNNs to DFT, by combining their approximation power with the physical nature of DFT, resulting in an accurate and data-efficient method.

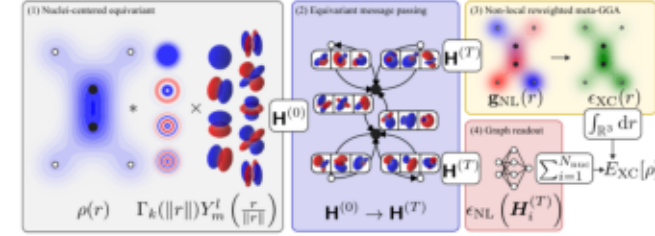


Figure 1: Illustration of **Equivariant Graph Exchange Correlation (EG-XC)**’s four components: (1) We obtain a finite **atomic-range** point cloud representation $\mathbf{H}^{(0)}$ by convolving the electron density ρ with radial filters $\Gamma_k: \mathbb{R}_+ \rightarrow \mathbb{R}$ and spherical harmonics $Y_m^l: \mathbb{R}^3 \rightarrow \mathbb{R}$ at the nuclear position. (2) The embeddings $\mathbf{H}^{(0)}$ are updated using equivariant message passing to obtain **molecular-range** effects in $\mathbf{H}^{(T)}$. (3) We define a non-local feature density $\mathbf{g}_{\text{NL}}: \mathbb{R}^3 \rightarrow \mathbb{R}^d$ from which we derive the exchange correlation density $\epsilon_{\text{XC}}: \mathbb{R}^3 \rightarrow \mathbb{R}$. (4) We add a graph readout of $\mathbf{H}^{(T)}$ to learn additional corrections. To obtain $E_{\text{XC}}[\rho]$, we integrate ϵ_{XC} and add it to the global graph readout.

4 EQUIVARIANT GRAPH EXCHANGE CORRELATION

With **Equivariant Graph Exchange Correlation (EG-XC)**, we propose an efficient *non-local* approximation to the unknown exchange-correlation functional E_{XC} that maps electron densities (positive integrable functions in $\mathbb{R}^3$) to scalar energies. To this end, EG-XC consists of four components as illustrated in Figure 1: (1) **Nuclei-centered equivariant embeddings** $\mathbf{H}^{(0)}$ compress the electron density ρ into a finite point cloud representation by equivariantly integrating the density around the nuclei **enabling atomic-range interactions**. (2) **Equivariant message passing** on this point cloud includes **molecular-range** information in $\mathbf{H}^{(T)}$. Based on the point cloud embeddings, we define E_{XC} as the sum of two readouts: (3) A **non-local reweighted meta-GGA** based on our non-local feature density $\mathbf{g}_{\text{NL}}: \mathbb{R}^3 \rightarrow \mathbb{R}^d$ and (4) a **graph readout** of the invariant nuclei-centered features $\mathbf{H}^{(T)}$.

$$E_{\text{XC}}[\rho] = \underbrace{\int_{\mathbb{R}^3} \rho(r) \gamma_{\text{NL}}(\mathbf{g}_{\text{NL}}(r), \mathbf{g}_{\text{meta-GGA}}(r)) \epsilon_{\text{meta-GGA}}(\mathbf{g}_{\text{meta-GGA}}(r)) dr}_{\text{non-local reweighted meta-GGA}} + \underbrace{\sum_{i=1}^{N_{\text{basis}}} \epsilon_{\text{NL}}(\mathbf{H}_i^{(T)})}_{\text{graph readout}}. \quad (8)$$

(1) **Nuclei-centered equivariant embeddings** $\mathbf{H}^{(0)}$ in $[\mathbb{R}^1 \times \dots \times \mathbb{R}^{2l_{\text{max}}+1}]^{N_{\text{basis}} \times d} := \mathbb{R}^{N_{\text{basis}} \times d}$ reduce the computational scaling by mapping the continuous density to a finite per-nucleus representation. Further, these enable using equivariant GNNs to learn **molecular-range** interactions efficiently. The positions of the nuclei $R_i \in \mathbb{R}^3$ naturally lend themselves as centroids for such embeddings as they represent peaks of the electron density (Kato, 1957). To perform this reduction, we evaluate a convolution of the electron density ρ with equivariant filters $S_k^l: \mathbb{R}^3 \rightarrow \mathbb{R}^{2l+1}$ at R_i :

$$H_{ik}^{(0)l} = (\rho_i * S_k^l)(R_i) \quad (9)$$

where $\rho_i: \mathbb{R}^3 \rightarrow \mathbb{R}_+$ is the soft-partitioned electron density associated with the i -th embedding

$$\rho_i(r) = \rho(r) \frac{\alpha_i(r)}{\sum_{j=1}^{N_{\text{basis}}} \alpha_j(r)}, \quad (10)$$

$$\alpha_i(r) = \exp\left(-\frac{\|r - R_i\|^2}{\lambda^2}\right) \quad (11)$$

with $\lambda \in \mathbb{R}_+$ being a free parameter. Such partitioning prevents overcounting if nuclei are close similar to quadrature methods (Becke, 1988). Like equivariant GNNs (Batzner et al., 2022), we define